

SQL Data Analysis Project –3 Decode Labs

Objective

The objective of this project is to analyze data using SQL queries and extract meaningful insights from a dataset. The project demonstrates the use of SQL commands such as SELECT, WHERE, ORDER BY, GROUP BY, COUNT, SUM, and AVG to retrieve, filter, sort, and summarize data efficiently.

Tools Used

- SQL Online Compiler / SQLite Online
- SQL Query Language
- Web Browser

Dataset Description

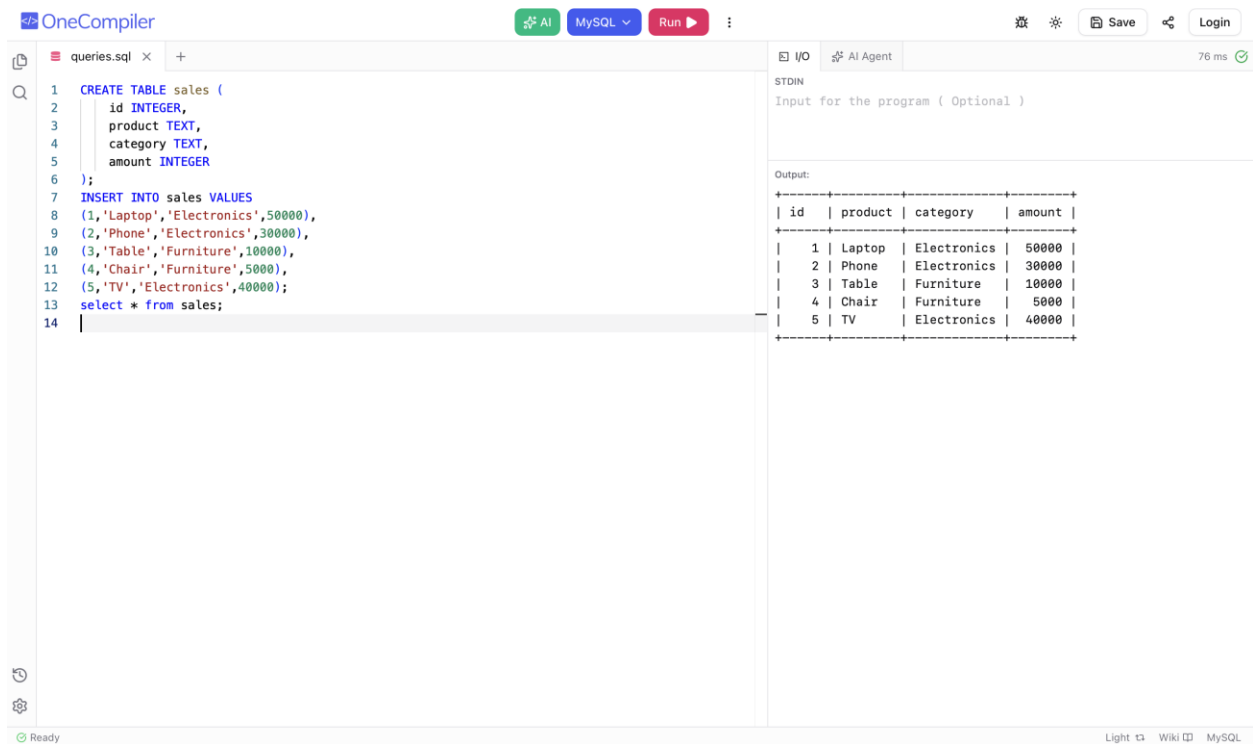
A sample sales dataset was created containing product information, product categories, and sales amounts. The dataset was used to perform various SQL operations and generate analytical insights.

SQL Queries Performed

1. SELECT Query – Display all records from the table.
2. WHERE Clause – Filter records based on sales amount.
3. ORDER BY Clause – Sort records in descending order.
4. COUNT Function – Count total records.

5. SUM Function – Calculate total sales amount.
6. AVG Function – Calculate average sales amount.
7. GROUP BY Clause – Group data by category and calculate total sales.

Screenshots



The screenshot shows the OneCompiler MySQL interface. The left pane contains a SQL query in a file named 'queries.sql'. The query creates a table named 'sales' with columns 'id' (INTEGER), 'product' (TEXT), 'category' (TEXT), and 'amount' (INTEGER). It then inserts five rows of data and finally selects all data from the 'sales' table.

```
1 CREATE TABLE sales (  
2   id INTEGER,  
3   product TEXT,  
4   category TEXT,  
5   amount INTEGER  
6 );  
7 INSERT INTO sales VALUES  
8 (1, 'Laptop', 'Electronics', 50000),  
9 (2, 'Phone', 'Electronics', 30000),  
10 (3, 'Table', 'Furniture', 10000),  
11 (4, 'Chair', 'Furniture', 5000),  
12 (5, 'TV', 'Electronics', 40000);  
13 select * from sales;  
14
```

The right pane shows the output of the query. It displays a table with 5 rows and 4 columns: id, product, category, and amount.

id	product	category	amount
1	Laptop	Electronics	50000
2	Phone	Electronics	30000
3	Table	Furniture	10000
4	Chair	Furniture	5000
5	TV	Electronics	40000

The interface also includes a top bar with 'OneCompiler', 'MySQL', 'Run', 'Save', and 'Login' buttons. The bottom status bar shows 'Ready' and 'Light Wiki MySQL'.

queries.sql

```
1 CREATE TABLE sales (  
2   id INTEGER,  
3   product TEXT,  
4   category TEXT,  
5   amount INTEGER  
6 );  
7 INSERT INTO sales VALUES  
8 (1,'Laptop','Electronics',50000),  
9 (2,'Phone','Electronics',30000),  
10 (3,'Table','Furniture',10000),  
11 (4,'Chair','Furniture',5000),  
12 (5,'TV','Electronics',40000);  
13  
14 SELECT * FROM sales  
15 WHERE amount > 10000;
```

I/O

AI Agent

77 ms

STDIN

Input for the program (Optional)

Output:

id	product	category	amount
1	Laptop	Electronics	50000
2	Phone	Electronics	30000
5	TV	Electronics	40000

Ready

Light Wiki MySQL

queries.sql

```
1 CREATE TABLE sales (  
2   id INTEGER,  
3   product TEXT,  
4   category TEXT,  
5   amount INTEGER  
6 );  
7 INSERT INTO sales VALUES  
8 (1,'Laptop','Electronics',50000),  
9 (2,'Phone','Electronics',30000),  
10 (3,'Table','Furniture',10000),  
11 (4,'Chair','Furniture',5000),  
12 (5,'TV','Electronics',40000);  
13  
14 SELECT * FROM sales  
15 ORDER BY amount DESC;
```

I/O

AI Agent

122 ms

STDIN

Input for the program (Optional)

Output:

id	product	category	amount
1	Laptop	Electronics	50000
5	TV	Electronics	40000
2	Phone	Electronics	30000
3	Table	Furniture	10000
4	Chair	Furniture	5000

Ready

Light Wiki MySQL

queries.sql

```
1 CREATE TABLE sales (  
2     id INTEGER,  
3     product TEXT,  
4     category TEXT,  
5     amount INTEGER  
6 );  
7 INSERT INTO sales VALUES  
8 (1,'Laptop','Electronics',50000),  
9 (2,'Phone','Electronics',30000),  
10 (3,'Table','Furniture',10000),  
11 (4,'Chair','Furniture',5000),  
12 (5,'TV','Electronics',40000);  
13  
14 SELECT COUNT(*) AS total_products  
15 FROM sales;
```

I/O

AI Agent

77 ms

STDIN

Input for the program (Optional)

Output:

```
+-----+  
| total_products |  
+-----+  
|          5 |  
+-----+
```

Ready

Light Wiki MySQL

queries.sql

```
1 CREATE TABLE sales (  
2     id INTEGER,  
3     product TEXT,  
4     category TEXT,  
5     amount INTEGER  
6 );  
7 INSERT INTO sales VALUES  
8 (1,'Laptop','Electronics',50000),  
9 (2,'Phone','Electronics',30000),  
10 (3,'Table','Furniture',10000),  
11 (4,'Chair','Furniture',5000),  
12 (5,'TV','Electronics',40000);  
13  
14 SELECT COUNT(*) AS total_products  
15 FROM sales;
```

I/O

AI Agent

109 ms

STDIN

Input for the program (Optional)

Output:

```
+-----+  
| total_products |  
+-----+  
|          5 |  
+-----+
```

Ready

Light Wiki MySQL

onecompiler.com

OneCompiler

queries.sql

```
1 CREATE TABLE sales (  
2   id INTEGER,  
3   product TEXT,  
4   category TEXT,  
5   amount INTEGER  
6 );  
7 INSERT INTO sales VALUES  
8 (1,'Laptop','Electronics',50000),  
9 (2,'Phone','Electronics',30000),  
10 (3,'Table','Furniture',10000),  
11 (4,'Chair','Furniture',5000),  
12 (5,'TV','Electronics',40000);  
13  
14 SELECT AVG(amount) AS average_sales  
15 FROM sales;
```

STDIN
Input for the program (Optional)

Output:

average_sales
27000.0000

Ready Light Wiki MySQL

onecompiler.com

OneCompiler

queries.sql

```
1 CREATE TABLE sales (  
2   id INTEGER,  
3   product TEXT,  
4   category TEXT,  
5   amount INTEGER  
6 );  
7 INSERT INTO sales VALUES  
8 (1,'Laptop','Electronics',50000),  
9 (2,'Phone','Electronics',30000),  
10 (3,'Table','Furniture',10000),  
11 (4,'Chair','Furniture',5000),  
12 (5,'TV','Electronics',40000);  
13  
14 SELECT category,  
15 SUM(amount) AS total_sales  
16 FROM sales  
17 GROUP BY category;
```

STDIN
Input for the program (Optional)

Output:

category	total_sales
Electronics	120000
Furniture	15000

Ready Light Wiki MySQL

Conclusion

This project helped in understanding the fundamentals of SQL data analysis. Various SQL operations were performed to retrieve, filter, sort, and aggregate data. The project demonstrated how SQL can be used to transform raw data into meaningful business insights, making it an essential skill for data analytics and decision-making.